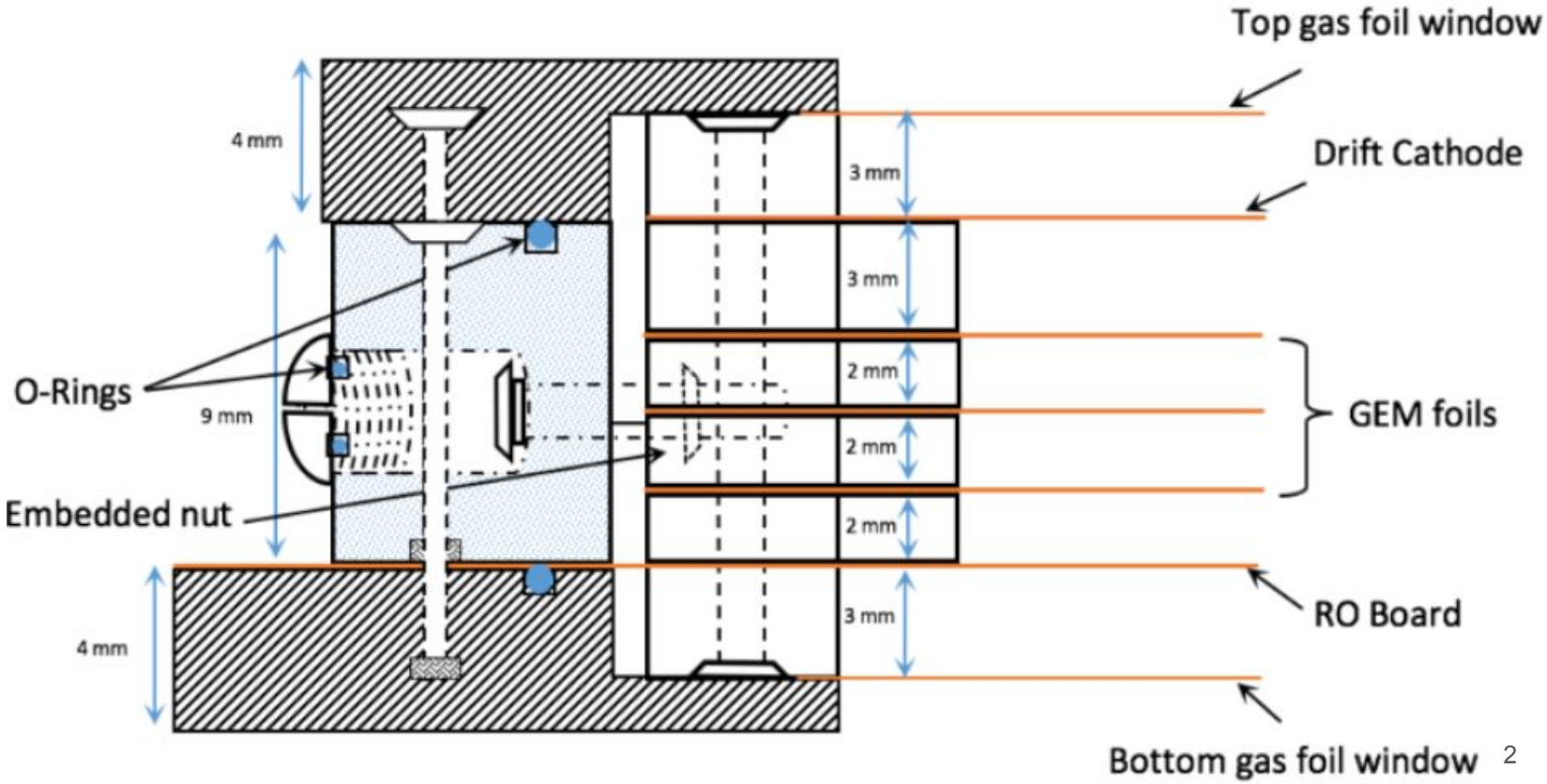
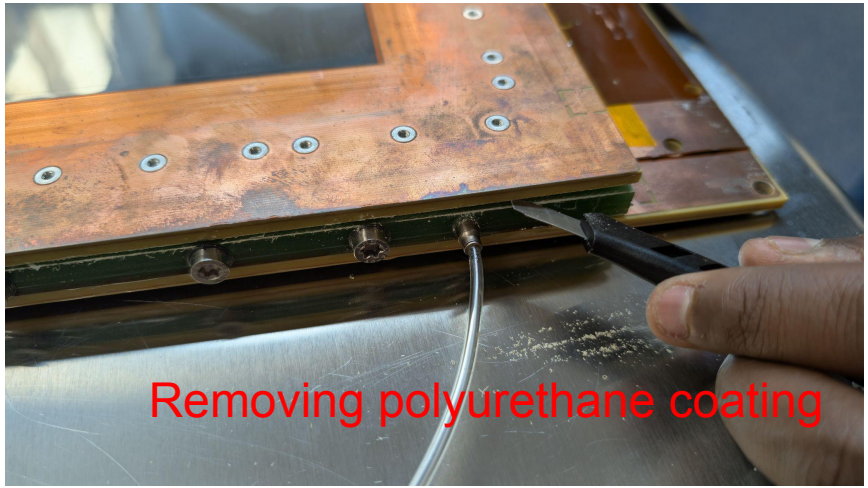
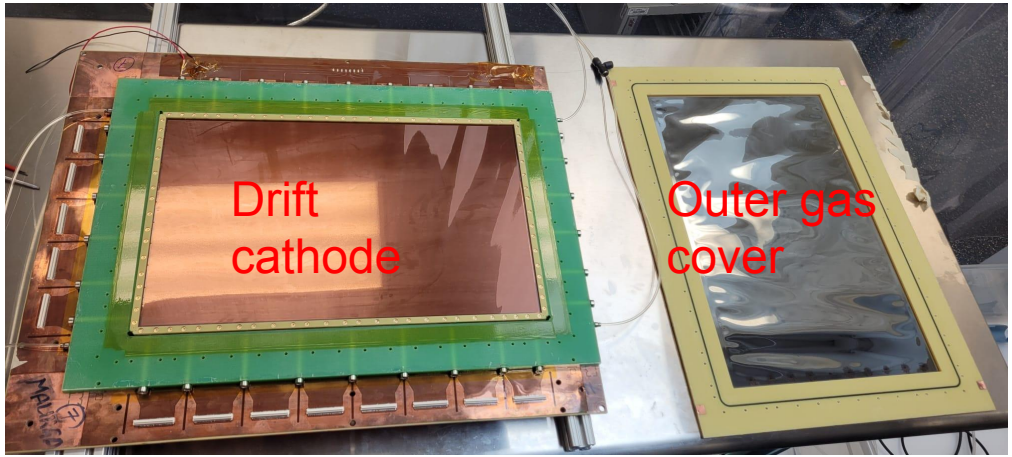
GEM Repair Update

(04/15/2026)

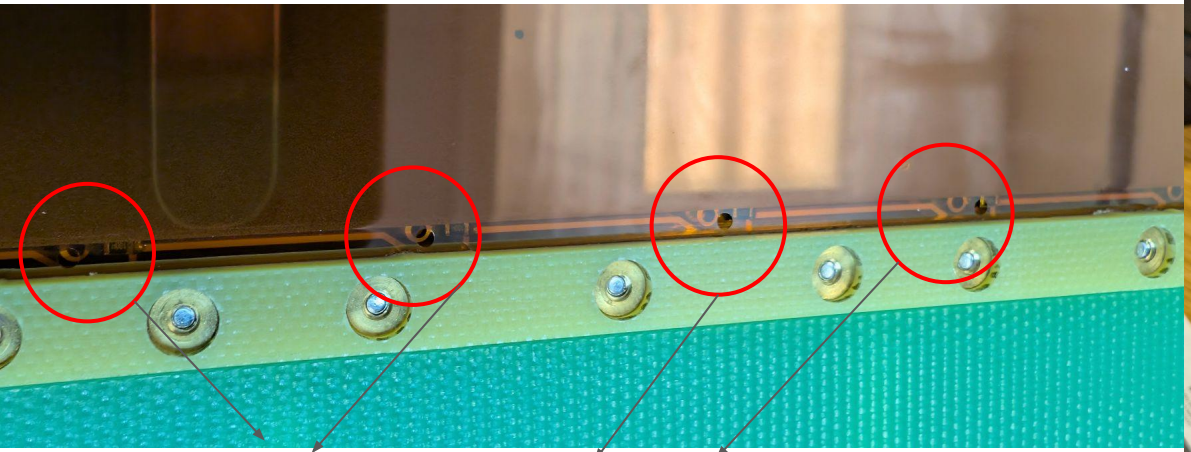
GEM Layout


Top gas foil window
Drift Cathode
4 mm
3 mm
3 mm
2 mm
2 mm
2 mm
3 mm
4 mm
9 mm
O-Rings
Embedded nut
GEM foils
RO Board
Bottom gas foil window


Removing polyurethane coating

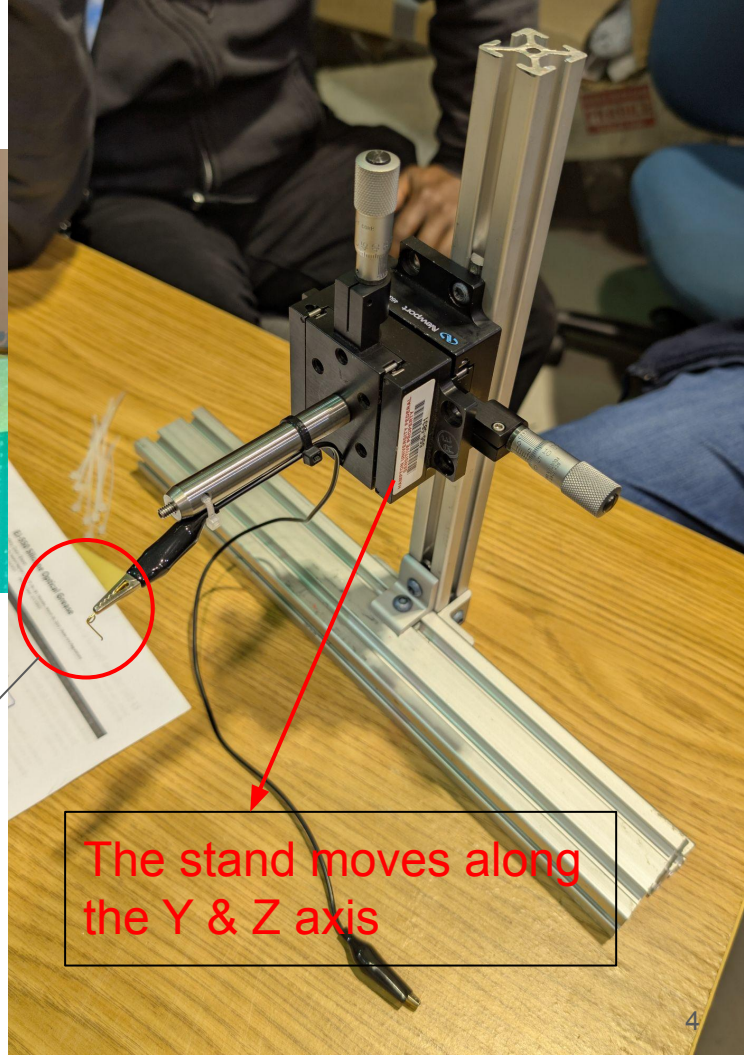

Drift
cathode
Outer gas
cover

Screws close to cathode line



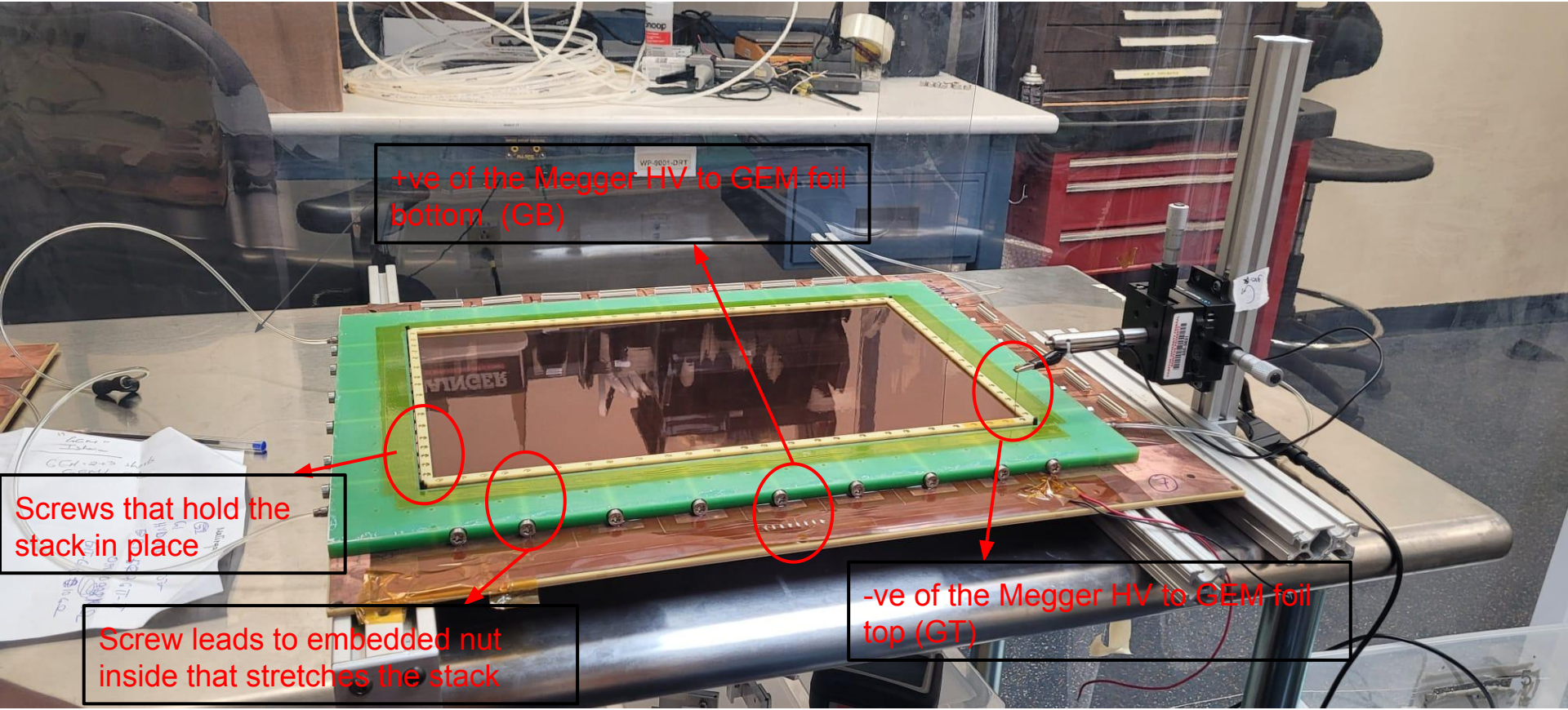
Each access point leads to a separate GEM sector.
Total of 10 sectors

Pin to probe the access points of G1, G2 & G3 top



The stand moves along the Y & Z axis

Complete Setup



Identifying Shorted GEM Foils..

- No short:
  - Measurements made after removing the HV divider
  - “OL” between GEM foil Top (GT) and foil Bottom (GB), measured with a multimeter.
  - 10 GΩ @ 50V, 20 GΩ @100V, 100GΩ @250V, between GT and GB, measured with the Megger.
  - Above measurements made **without** bypassing the protection resistor.
- Short:
  - ~1MΩ between GT & GB measured at the divider, **without** bypassing the protection resistor.
  - >2 MΩ between GT & GB, **after** bypassing the protection resistor, in every non-shorted sector (2 MΩ from 2 protection resistors + kΩ resistance from the short)
  - kΩ resistance between GT & GB in the shorted sector.
  - When bypassing the protection resistor, GB is at the HV divider solder point, and GT at any specific sector out of the 10.

GEM - Malinga / GEM # 7

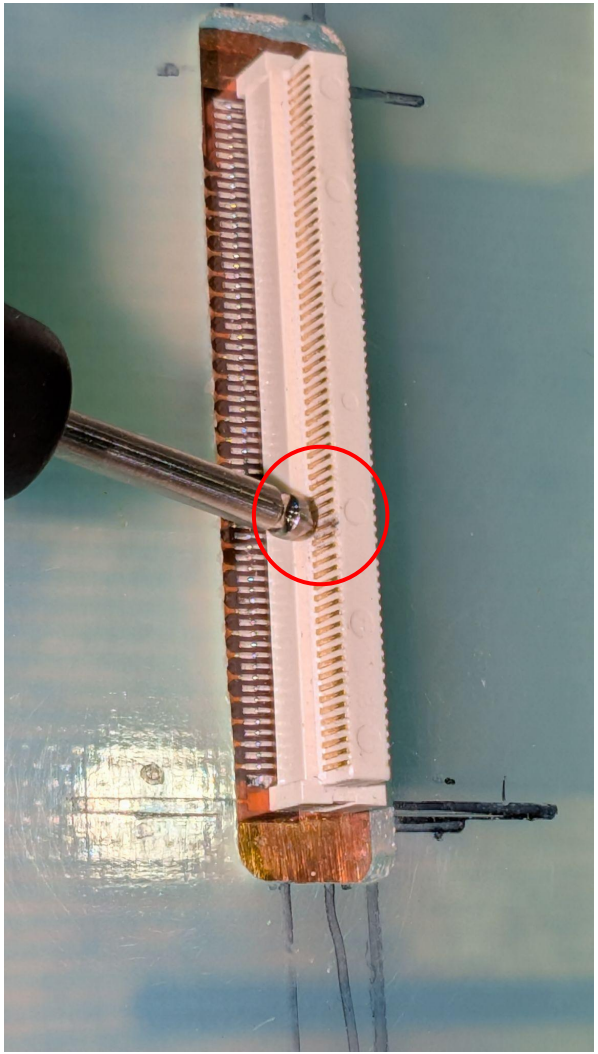
- Short in Gem Foil 2 (G2). G1 & G3, no short.
  - Applying <500V does not burn off the short. This causes the short to **migrate** to different sectors and even different foils.
  - After locating the shorted sector, apply 500V directly in short bursts to each foil after bypassing the protection resistor
  - Repeat until sparking reduces and eventually stops.
- Current Status
  - No shorts. “OL” between all GT & GB
  - Being flushed with Nitrogen to dry out the GEM.

GEM - Ishara / GEM # 1

  - Short in G2 & G3. No short in G1
  - 500V was not sufficient. Briefly apply 600V. Repeat until sparking reduces and eventually stops.
- Current Status
  - No shorts. “OL” between all GT & GB
  - Being flushed with Nitrogen to dry out the GEM.

GEM - Bishoy / GEM # 5

- Short between 2 channels of RO and the drift!
- Connected the strip buncher to the shorted Panasonic connector on the long side.
- 500V (already charged), no sparks. Voltage of the Megger drops to ~35V and 24 kOhm resistance.
- Same observation even at 1000V
- Unable to fix the short yet



Timeline of GEMs in e-Hall

Key: LG0 = Left bottom, LG1 = Left top, RG0 = Right bottom, RG1 = Right top.

Date / period	LG0	LG1	RG0	RG1	Spare / notes
Jun 25 (initial 4th GEM installed)	GEM-Ishara	GEM-Bishoy	GEM-Rui	GEM-Jesmin	Spare = Thir
Jul 9 (LG1 starts tripping/sparking)	GEM-Ishara	**GEM-Bishoy (trip/sparks)**	GEM-Rui	GEM-Jesmin	Spare = Thir
Jul 19 (RG1 also sparks)	GEM-Ishara	**GEM-Bishoy (sparks)**	GEM-Rui	**GEM-Jesmin (sparks)**	Spare = Thir
Jul 29 (major swap)	GEM-Ishara	**GEM-Thir (installed)**	GEM-Rui	**GEM-Malinga (installed)**	**Spare = Tanvi**
Aug 2025	GEM-Ishara	GEM-Thir	GEM-Rui	GEM-Malinga	Spare = Tanvi
Dec 10	**GEM-Ishara**	GEM-Thir	GEM-Rui (tripped)	GEM-Malinga	Spare = Tanvi
Dec 12 (LG0 fails)	**GEM-Ishara (failed)**	GEM-Thir	GEM-Rui	GEM-Malinga	Spare = Tanvi
Dec 14 (LG0 replaced)	**GEM-Tanvi (installed)**	GEM-Thir	GEM-Rui	GEM-Malinga	Spare = Jesmin
Jan 22, 2026 (RG1 fails)	GEM-Tanvi	GEM-Thir	GEM-Rui	**GEM-Malinga (failed)**	Spare = Jesmin
Jan 24 (RG1 replaced)	GEM-Tanvi	GEM-Thir	GEM-Rui	**GEM-Jesmin (installed)**	
Jan 24 (Entire right arm failed)	GEM-Tanvi	GEM-Thir	**GEM-Rui (failed)**	**GEM-Jesmin (failed)**	